

Ex7: Install and Run a Hadoop Single Node Cluster

Aim:

To set up and configure a **Hadoop Single Node Cluster** and run simple MapReduce applications like wordcount.

Requirements:

- Ubuntu 20.04 (or later)
- Java 8 or 11 (JDK)
- Hadoop 3.x (preferably 3.3.6)

Procedure :

Step 1: Install Java

```
sudo apt update
sudo apt install openjdk-11-jdk -y
java -version
```

Step 2: Add a New Hadoop User (optional)

```
sudo adduser hadoopuser
sudo usermod -aG sudo hadoopuser
su - hadoopuser
```

Step 3: Download and Extract Hadoop

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
tar -xzf hadoop-3.3.6.tar.gz
mv hadoop-3.3.6 hadoop
```

Step 4: Configure Environment Variables

Add to ~/.bashrc:

```
export HADOOP_HOME=~/.hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Then:

```
source ~/.bashrc
Edit $HADOOP_HOME/etc/hadoop/hadoop-env.sh:
```

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Step 5: Configure Hadoop for Single Node Cluster

core-site.xml

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

hdfs-site.xml

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
</configuration>
```

mapred-site.xml

First, create the file:

```
cp mapred-site.xml.template mapred-site.xml
```

Then edit:

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
</configuration>
```

yarn-site.xml

```
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
</configuration>
```

Step 6: Format the Namenode

```
hdfs namenode -format
```

Step 7: Start HDFS and YARN

```
start-dfs.sh
start-yarn.sh
To check:
```

```
jps
```

Expected output: NameNode, DataNode, ResourceManager, NodeManager, etc. **Run**

WordCount Example

Step 8: Create Input File in HDFS

```
hdfs dfs -mkdir /input
echo "Hello Hadoop Hello Cloud" > file.txt
hdfs dfs -put file.txt /input
```

Step 9: Run WordCount Job

```
cd $HADOOP_HOME
hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar wordcount /input
/output
```

♦ **Step 10: View Output**

```
hdfs dfs -cat /output/part-r-00000
Expected:
```

```
Cloud 1
Hadoop 1
Hello 2
```

Result:

Hadoop Single Node Cluster installed and simple MapReduce job wordcount executed successfully.

Viva Questions

1. What is HDFS and how does it work?
2. What are the roles of NameNode and DataNode?
3. What is the purpose of the wordcount example?
4. How does Hadoop handle large-scale data?
5. Difference between YARN and MapReduce?